

GENERIC REFINERY - GULF COAST 100 kbpsd Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU			COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-101 - Atmospheric crude fractionator			DOC. No.		CFU-000-ME-DS-001		
						REV / DATE		A / 2026-10-02		
						STATUS		Issued for review (FEED)		
						PREPARED		CLAUDE CODE CHK: - APP: -		
						SHEET		1		
GENERAL										
1	Tag / quantity		C-101 / 1			2	Service		Atmospheric crude fractionator	
3	Type		Trayed			4	Orientation		Vertical, skirt supported	
5	Code		ASME VIII Div.1 (U-stamp)			6	Wind / seismic		ASCE 7-16, 150 mph Exp. C / SDC B	
PROCESS & DESIGN CONDITIONS										
7	Operating pressure top / bottom		1.19 / 1.47 barg			8	Operating temperature top / bottom		134 / 361 C	
9	Design pressure (internal)		3.5 barg			10	Design pressure (external)		None (see note)	
11	Design temperature		390 C			12	MDMT		-5 C (site minimum)	
13	Corrosion allowance		3.0 mm			14	Joint efficiency / RT		1.0 / Full RT	
15	Hydrotest pressure (top / bottom)		5.74 / 9.54 barg			16	Test position		Vertical (field) - shop test of sections	
17	PWHT		Not required by code			18	Service fluid		Hydrocarbon + steam (sour)	
MATERIALS										
19	Shell / heads		SA-516-70			20	Cladding / lining		CS + 410S clad below tray 10; Monel 400 lined top ~	
21	Skirt		SA-516 Gr.70 (top 1 m same as shell)			22	Nozzle necks / forgings		SA-106 B / SA-105N	
23	Internals (trays/packing)		410S / 316L			24	Bolting (ext./int.)		SA-193 B7 / SA-194 2H ; int. SS	
25	Gaskets		Spiral-wound 316/graphite			26	Insulation		125 mm mineral wool (heat cons.)	
MECHANICAL DESIGN (CFU-000-ME-CAL-001)										
27	Inside diameter(s)		6900 / 4140 mm			28	Tangent-tangent height		36.00 m	
29	Shell thickness by course		Stripping section 18, Swage 30 deg 18, Flash zone ~			30	Top / bottom head (2:1 SE)		16 / 12 mm	
31	Skirt height / thickness / OD		6.5 m / 18 mm / 4176 mm			32	BTL elevation		EL 106.500	
33	Top of column		EL 144.225			34	Anchor bolts		32 x M48 F1554 Gr.55 on 4426 BCD	
35	Stiffening rings		None			36	Platforms / davit		8 platforms; davit on top head	
37	Fireproofing		Skirt inside & outside, 50 mm (API 2218)			38	Natural frequency / gust factor		1.00 Hz / 1.07	
INTERNALS										
39	Number of trays		41			40	Tray type		Valve trays (fixed/moving valves), 410S, 3 mm	
41	Tray spacing		610 / 760 mm			42	Passes		2 / 4	
43	Feed / draws		See nozzle schedule			44	Manways		7 x 24"	
WEIGHTS & LOADS										
45	Fabricated (shop) weight		261.7 t			46	Empty (installed)		337.4 t	
47	Operating		423.0 t			48	Hydrotest (full of water)		1554.5 t	
49	Wind base shear (strength)		756 kN			50	Wind base moment (strength)		19490 kNm	
NOZZLE SCHEDULE										
Mark	Qty	Size	Rating	Face	Service	Elevation		Orient.		
N1	1	36"	150#	RF	Feed - transfer line from H-101 (tangential, vapour horn)	EL 116.710		90 deg		
N2	1	36"	150#	RF	Overhead vapour to A-101	Top head		0 deg		
N3	1	8"	150#	RF	Reflux from P-103	EL 140.850		45 deg		
N4	1	8"	150#	RF	TPA return	EL 140.800		270 deg		
N5	1	14"	150#	RF	TPA draw to P-106 (draw sump)	EL 138.830		225 deg		
N8	1	10"	150#	RF	MPA return	EL 134.250		270 deg		
N9	1	16"	150#	RF	MPA draw to P-107 (draw sump)	EL 132.280		225 deg		
N12	1	8"	150#	RF	BPA return	EL 126.480		270 deg		
N13	1	16"	150#	RF	BPA draw to P-108 (draw sump)	EL 124.510		225 deg		
N6	1	12"	150#	RF	Kero draw to C-102	EL 134.410		135 deg		
N7	1	8"	150#	RF	Vapour return from C-102	EL 134.920		150 deg		
N10	1	12"	150#	RF	Diesel draw to C-103	EL 126.640		135 deg		
N11	1	8"	150#	RF	Vapour return from C-103	EL 127.150		150 deg		
N14	1	10"	150#	RF	Ago draw to C-104	EL 120.090		135 deg		
N15	1	6"	150#	RF	Vapour return from C-104	EL 120.600		150 deg		
N16	1	10"	150#	RF	Stripping steam (from H-101 SS coil)	EL 107.770		60 deg		
N17	1	16"	150#	RF	Atm. residue to P-112	Bottom head		0 deg		
N18	1	3"	150#	RF	Vent / steam-out	Top head		0 deg		
N19	4	8"	150#	RF	PSV-1001 relief (4 x T orifice)	Top head		0 deg		
N20A/B	2	3"	150#	RF	LT / LG bridle (bottom sump)	EL 106.800		300 deg		
N21	1	4"	150#	RF	Wash-zone overflash / slop connection (spare)	EL 118.260		200 deg		
M1	1	24"	150#	RF	Manway 24" (davit)	EL 141.500		0 deg		
M2	1	24"	150#	RF	Manway 24" (davit)	EL 135.475		0 deg		

GENERIC REFINERY - GULF COAST 100 kbpsd Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU	COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-101 - Atmospheric crude fractionator (continued)	DOC. No.	CFU-000-ME-DS-001		
		REV / DATE	A / 2026-10-02		
		STATUS	Issued for review (FEED)		
		PREPARED	CLAUDE CODE	CHK: -	APP: -
		SHEET	2		

NOZZLE SCHEDULE (CONT.)

Mark	Qty	Size	Rating	Face	Service	Elevation	Orient.
M3	1	24"	150#	RF	Manway 24" (davit)	EL 129.535	0 deg
M4	1	24"	150#	RF	Manway 24" (davit)	EL 122.375	0 deg
M5	1	24"	150#	RF	Manway 24" (davit)	EL 115.310	0 deg
M6	1	24"	150#	RF	Manway 24" (davit)	EL 109.895	0 deg
M7	1	24"	150#	RF	Manway 24" (davit)	EL 107.020	0 deg

HYDRAULIC SECTIONS (PROCESS SIZING)

Section	Trays	V (t/h)	L (t/h)	rho V	rho L	FLV	u flood/allow	D calc (m)	Passes
Top / TPA	1-3	229.2	268.1	5.025	611	0.106	1.12	4.60	2
Naphtha-kero fract.	4-9	334.5	206.6	5.607	651	0.057	1.02	5.51	4
MPA	11-13	620.0	438.5	7.526	599	0.079	0.94	6.74	4
Kero-diesel fract.	14-21	563.8	339.1	8.076	621	0.069	0.81	6.68	4
BPA	23-25	709.7	383.6	9.236	578	0.068	0.85	6.85	4
Diesel-AGO fract.	26-31	529.1	274.5	9.493	610	0.065	0.75	6.23	4
Wash zone	33-35	339.4	34.5	6.027	635	0.010	1.07	5.24	4
Stripping	36-41	15.3	256.9	1.596	706	0.796	0.82	2.47	2

NOTES

1. Thicknesses are FEED minimum nominal; fabricator to confirm by detailed calculation (cone junctions App. 1-5/1-8, nozzle reinforcement UG-37, local loads WRC 537).
2. Orientation angles are preliminary - final by piping layout (CFU-PI).
3. Steam-out vacuum: column not FV-rated unless noted - see CAL-001 S13.

GENERIC REFINERY - GULF COAST 100 kBPST Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU		COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-103 - Diesel side stripper		DOC. No. CFU-000-ME-DS-001			
				REV / DATE A / 2026-10-02			
				STATUS Issued for review (FEED)			
				PREPARED CLAUDE CODE CHK: - APP: -			
		SHEET 4					
GENERAL							
1	Tag / quantity	C-103 / 1	2	Service	Diesel side stripper		
3	Type	Trayed	4	Orientation	Vertical, skirt supported		
5	Code	ASME VIII Div.1 (U-stamp)	6	Wind / seismic	ASCE 7-16, 150 mph Exp. C / SDC B		
PROCESS & DESIGN CONDITIONS							
7	Operating pressure top / bottom	1.36 barg	8	Operating temperature top / bottom	302 C		
9	Design pressure (internal)	3.5 barg	10	Design pressure (external)	None (see note)		
11	Design temperature	330 C	12	MDMT	-5 C (site minimum)		
13	Corrosion allowance	6.0 mm	14	Joint efficiency / RT	1.0 / Full RT		
15	Hydrotest pressure (top / bottom)	4.77 / 5.76 barg	16	Test position	Vertical (field) - shop test of sections		
17	PWHT	Not required by code	18	Service fluid	Hydrocarbon + steam (sour)		
MATERIALS							
19	Shell / heads	SA-516-70	20	Cladding / lining	CS + 410S clad		
21	Skirt	SA-516 Gr.70 (top 1 m same as shell)	22	Nozzle necks / forgings	SA-106 B / SA-105N		
23	Internals (trays/packing)	410S / 316L	24	Bolting (ext./int.)	SA-193 B7 / SA-194 2H ; int. SS		
25	Gaskets	Spiral-wound 316/graphite	26	Insulation	100 mm mineral wool (heat cons.)		
MECHANICAL DESIGN (CFU-000-ME-CAL-001)							
27	Inside diameter(s)	1900 mm	28	Tangent-tangent height	9.20 m		
29	Shell thickness by course	Stripping section 12 mm	30	Top / bottom head (2:1 SE)	12 / 12 mm		
31	Skirt height / thickness / OD	4.5 m / 8 mm / 1924 mm	32	BTL elevation	EL 104.500		
33	Top of column	EL 114.175	34	Anchor bolts	16 x M24 F1554 Gr.55 on 2174 BCD		
35	Stiffening rings	None	36	Platforms / davit	2 platforms; davit on top head		
37	Fireproofing	Skirt inside & outside, 50 mm (API 2218)	38	Natural frequency / gust factor	4.26 Hz / 0.89		
INTERNALS							
39	Number of trays	6	40	Tray type	Valve trays (fixed/moving valves), 410S, 3 mm		
41	Tray spacing	610 mm	42	Passes	1		
43	Feed / draws	See nozzle schedule	44	Manways	2 x 24"		
WEIGHTS & LOADS							
45	Fabricated (shop) weight	14.4 t	46	Empty (installed)	26.5 t		
47	Operating	30.1 t	48	Hydrotest (full of water)	53.1 t		
49	Wind base shear (strength)	67 kN	50	Wind base moment (strength)	562 kNm		
NOZZLE SCHEDULE							
Mark	Qty	Size	Rating	Face	Service	Elevation	Orient.
N1	1	12"	150#	RF	Diesel feed from C-101 tray 22	EL 112.500	270 deg
N2	1	8"	150#	RF	Vapour return to C-101	Top head	0 deg
N3	1	6"	150#	RF	Stripping steam	EL 108.550	90 deg
N4	1	10"	150#	RF	Bottoms to P-110	Bottom head	0 deg
N5A/B	2	2"	150#	RF	LT / LG bridle	EL 104.700	120 deg
N6	1	3"	150#	RF	Vent / steam-out / PSV-1009 (common)	Top head	0 deg
M1	1	24"	150#	RF	Manway 24"	EL 112.950	0 deg
M2	1	24"	150#	RF	Manway 24"	EL 107.800	0 deg
NOTES							
1. Thicknesses are FEED minimum nominal; fabricator to confirm by detailed calculation (cone junctions App. 1-5/1-8, nozzle reinforcement UG-37, local loads WRC 537).							
2. Orientation angles are preliminary - final by piping layout (CFU-PI).							
3. Steam-out vacuum: column not FV-rated unless noted - see CAL-001 S13.							

GENERIC REFINERY - GULF COAST 100 kbpsd Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU	COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-105 - Naphtha stabiliser (debutaniser)	DOC. No.	CFU-000-ME-DS-001	
		REV / DATE	A / 2026-10-02	
		STATUS	Issued for review (FEED)	
		PREPARED	CLAUDE CODE	CHK: - APP: -
		SHEET	6	

GENERAL

1	Tag / quantity	C-105 / 1	2	Service	Naphtha stabiliser (debutaniser)
3	Type	Trayed	4	Orientation	Vertical, skirt supported
5	Code	ASME VIII Div.1 (U-stamp)	6	Wind / seismic	ASCE 7-16, 150 mph Exp. C / SDC B

PROCESS & DESIGN CONDITIONS

7	Operating pressure top / bottom	10.8 barg	8	Operating temperature top / bottom	79 / 203 C
9	Design pressure (internal)	13.0 barg	10	Design pressure (external)	None (see note)
11	Design temperature	235 C	12	MDMT	-5 C (site minimum)
13	Corrosion allowance	3.0 mm	14	Joint efficiency / RT	1.0 / Full RT
15	Hydrotest pressure (top / bottom)	16.90 / 18.77 barg	16	Test position	Vertical (field) - shop test of sections
17	PWHT	Yes (HIC / sour, NACE MR0103)	18	Service fluid	LPG / naphtha (sour, H2S)

MATERIALS

19	Shell / heads	SA-516-70	20	Cladding / lining	CS (HIC-resistant plate, NACE MR0103)
21	Skirt	SA-516 Gr.70 (top 1 m same as shell)	22	Nozzle necks / forgings	SA-106 B / SA-105N
23	Internals (trays/packing)	410S	24	Bolting (ext./int.)	SA-193 B7 / SA-194 2H ; int. SS
25	Gaskets	Spiral-wound 316/graphite	26	Insulation	75 mm mineral wool (heat cons.)

MECHANICAL DESIGN (CFU-000-ME-CAL-001)

27	Inside diameter(s)	2100 mm	28	Tangent-tangent height	18.00 m
29	Shell thickness by course	Shell 14 mm	30	Top / bottom head (2:1 SE)	16 / 16 mm
31	Skirt height / thickness / OD	6.0 m / 8 mm / 2128 mm	32	BTL elevation	EL 106.000
33	Top of column	EL 124.525	34	Anchor bolts	20 x M30 F1554 Gr.55 on 2378 BCD
35	Stiffening rings	None	36	Platforms / davit	4 platforms; davit on top head
37	Fireproofing	Skirt inside & outside, 50 mm (API 2218)	38	Natural frequency / gust factor	1.66 Hz / 0.88

INTERNALS

39	Number of trays	19	40	Tray type	Valve trays (fixed/moving valves), 410S, 3 mm
41	Tray spacing	610 mm	42	Passes	1 / 2
43	Feed / draws	See nozzle schedule	44	Manways	3 x 24"

WEIGHTS & LOADS

45	Fabricated (shop) weight	27.2 t	46	Empty (installed)	48.1 t
47	Operating	53.9 t	48	Hydrotest (full of water)	110.7 t
49	Wind base shear (strength)	138 kN	50	Wind base moment (strength)	1992 kNm

NOZZLE SCHEDULE

Mark	Qty	Size	Rating	Face	Service	Elevation	Orient.
N1	1	8"	300#	RF	Feed (tray 13)	EL 114.980	90 deg
N2	1	8"	300#	RF	Overhead vapour to A-106	Top head	0 deg
N3	1	3"	300#	RF	Reflux	EL 122.350	270 deg
N4	1	16"	300#	RF	Reboiler feed to E-116 (kettle)	Bottom head	0 deg
N5	1	10"	300#	RF	Reboiler return from E-116 (kettle)	EL 110.220	120 deg
N6	1	12"	300#	RF	Stabilised naphtha (bottoms) to E-114/C-106	EL 106.300	200 deg
N7	1	6"	300#	RF	PSV-1005 relief (R)	Top head	0 deg
N8A/B	2	2"	300#	RF	LT / LG bridle	EL 106.200	300 deg
N9	1	2"	300#	RF	Vent / steam-out	Top head	0 deg
M1	1	24"	300#	RF	Manway 24"	EL 123.000	0 deg
M2	1	24"	300#	RF	Manway 24"	EL 114.985	0 deg
M3	1	24"	300#	RF	Manway 24"	EL 109.520	0 deg

NOTES

1. Thicknesses are FEED minimum nominal; fabricator to confirm by detailed calculation (cone junctions App. 1-5/1-8, nozzle reinforcement UG-37, local loads WRC 537).
2. Orientation angles are preliminary - final by piping layout (CFU-PI).
3. Steam-out vacuum: column not FV-rated unless noted - see CAL-001 S13.

GENERIC REFINERY - GULF COAST 100 kbpsd Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: CDU	COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-106 - Naptha splitter	DOC. No.	CFU-000-ME-DS-001	
		REV / DATE	A / 2026-10-02	
		STATUS	Issued for review (FEED)	
		PREPARED	CLAUDE CODE	CHK: - APP: -
		SHEET	7	

GENERAL

1	Tag / quantity	C-106 / 1	2	Service	Naphtha splitter
3	Type	Trayed	4	Orientation	Vertical, skirt supported
5	Code	ASME VIII Div.1 (U-stamp)	6	Wind / seismic	ASCE 7-16, 150 mph Exp. C / SDC B

PROCESS & DESIGN CONDITIONS

7	Operating pressure top / bottom	1.1 barg	8	Operating temperature top / bottom	84 / 152 C
9	Design pressure (internal)	3.5 barg	10	Design pressure (external)	None (see note)
11	Design temperature	185 C	12	MDMT	-5 C (site minimum)
13	Corrosion allowance	3.0 mm	14	Joint efficiency / RT	1.0 / Full RT
15	Hydrotest pressure (top / bottom)	4.55 / 7.60 barg	16	Test position	Vertical (field) - shop test of sections
17	PWHT	Not required by code	18	Service fluid	Hydrocarbon + steam (sour)

MATERIALS

19	Shell / heads	SA-516-70	20	Cladding / lining	CS
21	Skirt	SA-516 Gr.70 (top 1 m same as shell)	22	Nozzle necks / forgings	SA-106 B / SA-105N
23	Internals (trays/packing)	410S / 316L	24	Bolting (ext./int.)	SA-193 B7 / SA-194 2H ; int. SS
25	Gaskets	Spiral-wound 316/graphite	26	Insulation	75 mm mineral wool (heat cons.)

MECHANICAL DESIGN (CFU-000-ME-CAL-001)

27	Inside diameter(s)	3100 mm	28	Tangent-tangent height	29.50 m
29	Shell thickness by course	Shell 12 mm	30	Top / bottom head (2:1 SE)	12 / 12 mm
31	Skirt height / thickness / OD	7.5 m / 12 mm / 3124 mm	32	BTL elevation	EL 107.500
33	Top of column	EL 137.775	34	Anchor bolts	24 x M42 F1554 Gr.55 on 3374 BCD
35	Stiffening rings	None	36	Platforms / davit	6 platforms; davit on top head
37	Fireproofing	Skirt inside & outside, 50 mm (API 2218)	38	Natural frequency / gust factor	1.16 Hz / 0.88

INTERNALS

39	Number of trays	38	40	Tray type	Valve trays (fixed/moving valves), 410S, 3 mm
41	Tray spacing	610 mm	42	Passes	2
43	Feed / draws	See nozzle schedule	44	Manways	5 x 24"

WEIGHTS & LOADS

45	Fabricated (shop) weight	69.1 t	46	Empty (installed)	108.7 t
47	Operating	129.7 t	48	Hydrotest (full of water)	334.1 t
49	Wind base shear (strength)	294 kN	50	Wind base moment (strength)	6308 kNm

NOZZLE SCHEDULE

Mark	Qty	Size	Rating	Face	Service	Elevation	Orient.
N1	1	8"	150#	RF	Feed (tray 21)	EL 123.100	90 deg
N2	1	20"	150#	RF	Overhead vapour to A-107	Top head	0 deg
N3	1	6"	150#	RF	Reflux	EL 135.350	270 deg
N4	1	16"	150#	RF	Reboiler feed to E-117 (thermosyphon)	Bottom head	0 deg
N5	1	20"	150#	RF	Reboiler return from E-117 (thermosyphon)	EL 111.630	120 deg
N6	1	10"	150#	RF	Bottoms to P-117A/B	EL 107.800	200 deg
N7	1	8"	150#	RF	PSV-1007 relief (T)	Top head	0 deg
N8A/B	2	2"	150#	RF	LT / LG bridle	EL 107.700	300 deg
N9	1	2"	150#	RF	Vent / steam-out	Top head	0 deg
M1	1	24"	150#	RF	Manway 24"	EL 136.000	0 deg
M2	1	24"	150#	RF	Manway 24"	EL 129.205	0 deg
M3	1	24"	150#	RF	Manway 24"	EL 123.105	0 deg
M4	1	24"	150#	RF	Manway 24"	EL 117.005	0 deg
M5	1	24"	150#	RF	Manway 24"	EL 110.930	0 deg

NOTES

1. Thicknesses are FEED minimum nominal; fabricator to confirm by detailed calculation (cone junctions App. 1-5/1-8, nozzle reinforcement UG-37, local loads WRC 537).
2. Orientation angles are preliminary - final by piping layout (CFU-PI).
3. Steam-out vacuum: column not EV-rated unless noted - see CAL-001 S13.

GENERIC REFINERY - GULF COAST 100 kBPSD Crude & Vacuum Distillation Unit Stage: FEED - Discipline: MECHANICAL Unit area: VDU	COLUMN DATASHEET ASME VIII Div.1 - columns & internals C-201 - Vacuum column (wet, packed) (continued)	DOC. No.	CFU-000-ME-DS-001		
		REV / DATE	A / 2026-10-02		
		STATUS	Issued for review (FEED)		
		PREPARED	CLAUDE CODE	CHK: -	APP: -
		SHEET	9		

HYDRAULIC SECTIONS (PROCESS SIZING)									
Section	Trays	V (t/h)	L (t/h)	rho V	rho L	FLV	u flood/allow	D calc (m)	Passes
Bed 1 - LVGO PA	-	10.4	82.5	0.037	801	-	16.26	2.49	-
Bed 2 - LVGO/HVGO fract.	-	87.2	34.0	0.176	734	-	7.11	4.97	-
Bed 3 - HVGO PA	-	177.6	161.5	0.283	680	-	5.39	6.41	-
Bed 4 - Wash	-	151.9	12.6	0.307	711	-	5.77	5.50	-
Flash zone	-	151.9	107.9	0.327	741	-	5.23	5.60	-
Stripping	-	5.0	108.3	0.041	757	0.157	11.12	2.49	2

NOTES

1. Thicknesses are FEED minimum nominal; fabricator to confirm by detailed calculation (cone junctions App. 1-5/1-8, nozzle reinforcement UG-37, local loads WRC 537).
2. Orientation angles are preliminary - final by piping layout (CFU-PI).
3. Steam-out vacuum: column not FV-rated unless noted - see CAL-001 S13.